

Ultra-broadband, Ultra-Dense 2D-Array Optical I/O Interface Enabled by 3D Nanoprinting

Huiyu Huang⁽¹⁾, Zhitian Shi⁽²⁾, Chunhui Yao^(1,3), Richard Penty⁽¹⁾, Qixiang Cheng^(1,3,*)

⁽¹⁾ Electrical Engineering Division, Department of Engineering, University of Cambridge, Cambridge, United Kingdom, CB3 0FA, qc223@cam.ac.uk

⁽²⁾ Southeast University, School of Electrical Science & Engineering, Nanjing 210096, Jiangsu, Peoples Republic of China

⁽³⁾ GlitterinTech Ltd, Cambridge, United Kingdom, CB3 0EJ

Abstract We demonstrate a 3D-nanoprinted high-efficiency, ultra-dense 2D-array optical I/O interface for photonic packaging, integrating 16 channels within $160 \times 120 \mu\text{m}^2$. Experiments show stable 1–2 dB channel losses, with rigorous simulations indicating a potential reduction to 0.4 dB. ©2026 The Author(s)

Introduction

Optical input/output (I/O) is becoming a key enabling technology for next-generation computing and communication systems, driven by the need for higher bandwidth density, lower energy consumption, and shorter electrical interconnect reach^[1]. As photonic integration advances in processors, switches, and chiplet-based architectures, optical interfaces must support increasing channel counts within limited package footprints^[2]. In this context, dense optical I/O is no longer determined solely by per-channel coupling efficiency but also by interface pitch, scalability, assembly tolerance, and array-level uniformity^[3]. However, many existing chip-coupling approaches remain optimized primarily for single-channel operation, making them difficult to translate directly into practical dense optical I/O interfaces.

A central challenge lies in the pitch-constrained nature of optical I/O integration. External optical ecosystems are commonly based on standardized fibre arrays, whereas chip-side photonic interfaces tend to pursue much denser channel arrangements. As a result, the optical interface must not only provide efficient coupling, but also accommodate constraints associated with channel spacing,

footprint, crosstalk, and packaging compatibility. These requirements become even more critical as the number of parallel channels increases, particularly when simplified or passive assembly is desired. Therefore, there is a pressing need for compact, scalable coupling structures that operate under practically relevant pitch conditions while maintaining robust array-level performance.

In this work, we present a 4×4 optical I/O coupling interface designed for parallel chip-scale interconnection with an x-pitch of $40 \mu\text{m}$ and a y-pitch of $30 \mu\text{m}$. Rather than simply extending a single-channel coupler, the structure is developed as a package-aware optical interface building block for dense optical I/O. By co-designing the optical collimation and coupling regions, the device enables sixteen-channel integration within a compact footprint of only $160 \times 120 \mu\text{m}^2$, while maintaining low loss and high repeatability. For each channel, simulations predict a coupling loss as low as 0.4 dB over a broad wavelength range from 1200 nm to 1800 nm thanks to the free-space propagating nature. The loss of the 4×4 optical interface was experimentally measured channel by channel, with all channels exhibiting losses in the range of 1–2 dB. The additional loss observed in the

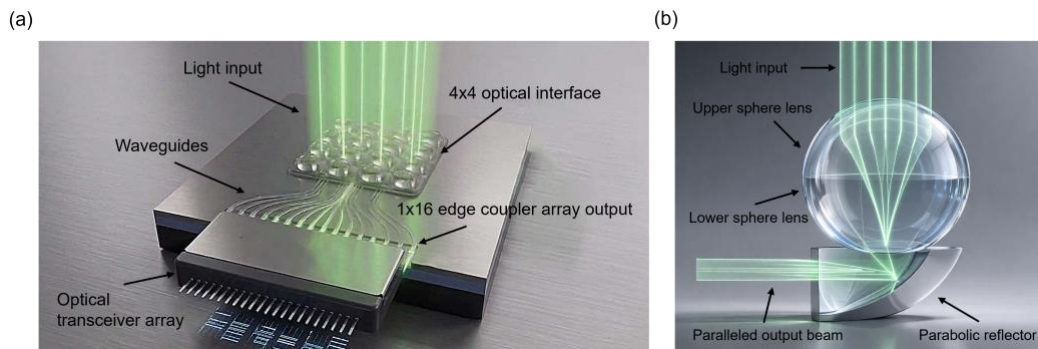


Fig. 1 (a) Schematic view of the 4×4 optical I/O interface. (b) Schematic of the 3D-nanoprinted optical coupler in one pixel. The sphere lens collects the light from free-space and the parabolic reflector collimate and control the mode size of the output beam.

experiments is mainly attributed to misalignment and fabrication errors. With further refinement of the fabrication recipe and longer fabrication time, the loss is expected to be further reduced. By changing the photonic chip design, the pitch size can be further reduced. These results demonstrate the viability of the proposed 4×4 structure as a scalable parallel optical I/O interface and highlight its potential for future dense photonic packaging architectures.

Design and simulations

The proposed optical I/O interface was developed from an array-level perspective to support parallel coupling under defined pitch constraints. The overall structure was divided into a designated printing area and a 2D input region, allowing the optical interface to be organized in a scalable array format. To facilitate dense integration, the array geometry was constructed using a pixelated layout strategy, which provides a regular framework for channel placement and subsequent 3D printing. Each pixel has a footprint of $40\ \mu\text{m} \times 30\ \mu\text{m}$.

For each optical channel, the coupling unit was designed using a double semi-spherical input lens combined with a parabolic reflector. This configuration was chosen to improve light

upper sphere was fixed at $30\ \mu\text{m}$, as this dimension provides good coverage of the pixel area while maintaining efficient light collection. Based on this configuration, a series of simulations was carried out by fixing the upper radius at $30\ \mu\text{m}$ and varying the lower radius from $10\ \mu\text{m}$ to $40\ \mu\text{m}$. This parametric sweep was used to identify the most suitable geometry for efficient light transfer and achieve the smallest mode output for the following parabolic reflector. According to the simulation result shown in Fig. 2, the lower radius is chosen to be $35\ \mu\text{m}$. The geometry of the parabolic reflector comes from our previous work, which has been validated in mode size conversion^[4]. For the entire device, the simulation results show that the lowest loss can reach 0.4 dB over a broadband wavelength range from 1200 nm to 1800 nm, benefiting from the free-space propagation nature of the device.

Fabrication

The 4×4 optical I/O interface was fabricated using a commercial direct laser writing system (Nanoscribe Professional GT2) equipped with a 25× objective lens. The structure was printed using IP-series photoresist, with the layout defined according to the designated array area and the corresponding 2D input region. To

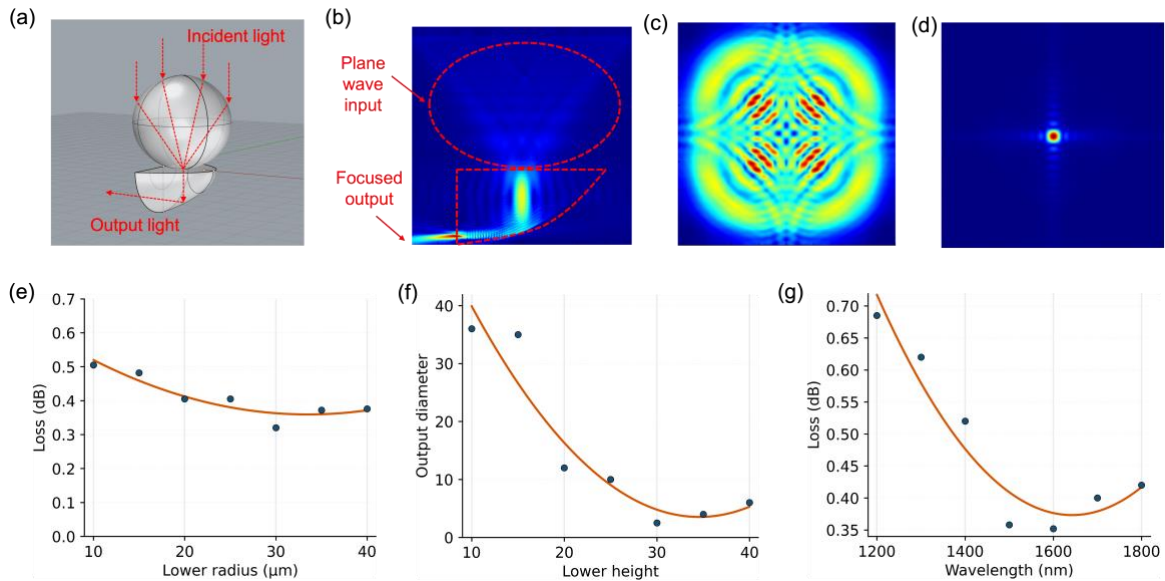


Fig. 2 (a) Schematic view of the coupler in one pixel. (b) Propagation profile of the light inside the 3D-nanoprinted optical coupler. The coupler collects the plane wave input and output the focused light beam. The beam is received by a waveguide. (c), (d) output profiles under different lower lens geometries. (c) $15\ \mu\text{m}$ and (d) $30\ \mu\text{m}$. (e) Loss of the sphere lens under different radius of lower sphere. (f) Output beam diameter of the sphere lens under different radius of lower sphere. (g) Simulated broadband performance of the optical coupler.

collection and beam redirection while preserving sufficient structural simplicity for array implementation. The semi-spherical lens maximizes free-space light collection, while the reflector reshapes the mode size to match that of the waveguide. In the design, the radius of the

accommodate the parallel optical interface within a compact footprint, the full device was divided into several functional parts, including the optical coupling section and the supporting mechanical section, which were integrated into a unified printing file for array fabrication.

To balance fabrication efficiency and optical quality, different printing conditions were applied to different parts of the structure. A coarse printing recipe was adopted for the collimating sphere, while a finer recipe was used for the parabolic reflector to achieve smoother interfaces and reduce scattering loss. This partitioned strategy is particularly beneficial for the present array structure, as it allows multiple channels to be fabricated within a reasonable writing time while maintaining the optical quality required for dense optical I/O operation. After recipe optimization, the fabrication time of the optical interface can be controlled within 1 hour.

Results and Discussions

The multi-channel coupling tests were performed on the complete 4×4 structure. Each of the sixteen channels was tested individually, and the array performance was evaluated across multiple chips. According to the measurements, the losses of all 16 channels on four different chips fall within the range of 1–2 dB. The measured responses remain stable from channel to channel, with no obvious degradation in operation. These

consistent across repeated samples and array positions. Such behaviour is essential for optical I/O applications, where practical usefulness depends not only on the performance of an individual coupler, but also on the robustness and reproducibility of the full interface. These results therefore support the feasibility of the proposed 4×4 structure as a stable and scalable optical I/O building block.

Conclusions

In conclusion, we demonstrate a 3D-nanoprinted 4×4 optical I/O interface for dense and scalable photonic packaging. The proposed structure combines a pixelated array layout with semi-spherical input lenses and parabolic reflectors, enabling sixteen-channel integration within a compact $160 \times 120 \mu\text{m}^2$ footprint under $40 \mu\text{m} \times 30 \mu\text{m}$ pitch constraints. Simulations indicate broadband low-loss operation, with the optimized design reaching a coupling loss as low as 0.4 dB from 1200 nm to 1800 nm. Experimentally, channel-by-channel measurements across multiple chips show losses within 1–2 dB,

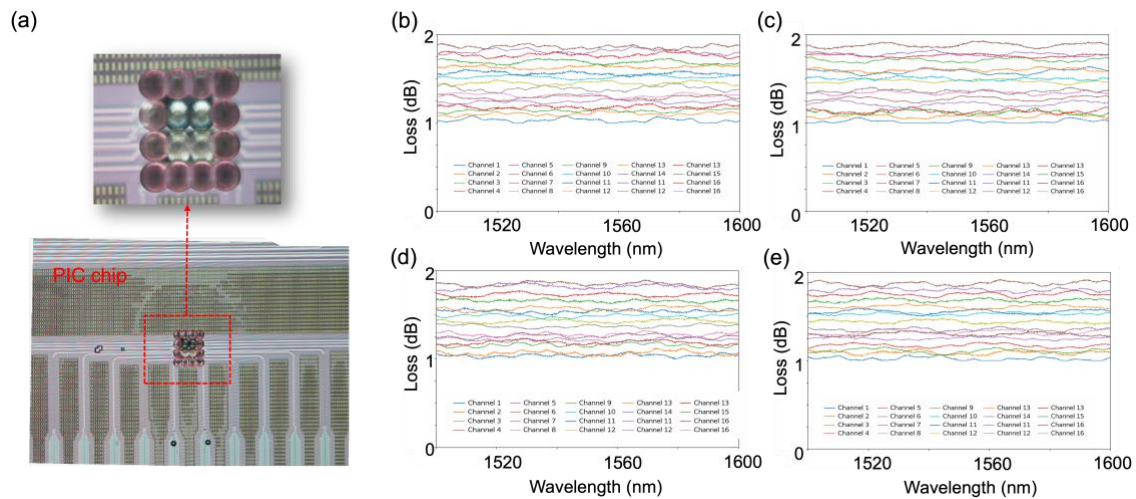


Fig. 3 (a) on-chip printed coupler array. (b)-(d) Loss tests of the 4×4 optical I/O interface fabricated on 4 different chips. The test is conducted channel by channel.

results suggest that the proposed structure preserves its coupling functionality in array integration and supports parallel optical I/O operation within the designed footprint. The higher loss compared with the simulation is mainly attributed to the trade-off made in the fabrication recipe. With further refinement of the printing conditions, the loss can be further reduced.

Importantly, repeated measurements across multiple chips and channels indicate good stability and repeatability of the proposed interface. The channel-by-channel testing shows that the device response is not dominated by isolated outliers, but instead remains broadly

confirming stable array-level performance and good repeatability. Although the measured loss is higher than the simulated value due to fabrication and alignment limitations, the results verify the feasibility of the proposed architecture as a practical optical I/O building block. Notably, the pitch size is primarily determined by the photonic chip itself, suggesting that the channel pitch could be further reduced by using different chip platforms. This work provides a promising route toward compact, high-density, and manufacturable optical interfaces for future photonic interconnect and packaging systems.

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